Quantum Machine Learning Foundations & Noise Dynamics

A Rigorous Mathematical Compendium of Equations, Derivations, and Interpretations

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Executive Introduction & Theoretical Scope

This document serves as the foundational mathematical reference manual for the project **"Dual-Noise Regularized Quantum Graph Neural Networks (QGNNs) for High-Throughput Molecular Property Prediction"**. It provides explicit LaTeX typesetting for all 16 core equations governing physical hardware noise regularization, loss landscape curvature balancing, Hilbert-space margin bounds, Log-Ratio Probability (LRP) gradient pumping, and classical Fourier surrogacy. Each equation is paired with a **term-by-term breakdown table**, a **step-by-step mathematical derivation**, and a **physical mechanism illustration**.

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1 Pillar 1: Physical Quantum Hardware Noise & Stochastic Capacity Regularization

Equation 1.1: Microscopic Bernoulli Noise Channel Insertion

$$\mathcal{E}_{j,z_j}(\rho) = (1 - z_j)\rho + z_j\mathcal{N}_j(\rho), \quad \text{where } z_j \sim \text{Ber}(\beta)$$

Table 1: Term-by-Term Interpretation for Equation 1.1

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
ρ	$\rho \in \mathcal{S}(\mathcal{H})$, $\text{Tr}(\rho) = 1$	Input density matrix representing the quantum state at gate location j.
z_j	$z_j \in \{0, 1\}$	Bernoulli random variable acting as a stochastic noise trigger switch.
β	$\beta \in [0, 0.5]$	Microscopic physical error rate per gate location across the QPU.
$\mathcal{N}_j(\rho)$	Completely Positive Trace-Preserving (CPTP) map	Quantum noise superoperator (e.g., bit-flip, phase-flip, depolarizing).
$\mathcal{E}_{j,z_j}(\rho)$	Conditional quantum channel	Realized noisy quantum state conditioned on the random noise instance z_j.

Step-by-Step Derivation & Mathematical Proof

Derivation: Consider a quantum circuit consisting of g discrete gate operations. At each gate location $j \in \{1, 2, \ldots, g\}$, environmental interaction or imperfect pulse control inserts an unwanted noise process with independent probability β. We model this insertion using a Bernoulli random variable z_j with probability mass function $P(z_j = 1) = \beta$ and $P(z_j = 0) = 1 - \beta$.
When $z_j = 0$, the state evolves under the ideal identity map $\text{Id}(\rho) = \rho$. When $z_j = 1$, the ideal gate is followed by the noise channel $\mathcal{N}_j(\rho)$. Combining these mutually exclusive outcomes linearly yields the stochastic quantum channel superoperator $\mathcal{E}_{j,z_j}(\rho) = (1 - z_j)\rho + z_j\mathcal{N}_j(\rho)$.

Physical Mechanism & Intuitive Interpretation

Physical Illustration: Gate-level physical noise acts as a stochastic branching process. Rather than deterministically destroying the quantum state, physical noise creates a statistical ensemble of clean and noisy trajectory execution branches across the quantum processor.

Equation 1.2: Counting Function & Order-m Conditional Predictor

$$c(z) = \sum_{j=1}^{g} z_j, \quad \psi_m(\rho_0; \theta) = \mathbb{E}_z \left[\phi_z(\rho_0; \theta) \mid c(z) = m\right]$$

Table 2: Term-by-Term Interpretation for Equation 1.2

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$z = (z_1, \ldots, z_g)$	$z \in \{0,1\}^g$	Microscopic noise realization vector across all g gate locations.
$c(z)$	$c(z) \in \{0, 1, \ldots, g\}$	Total noise occurrence counting function across the entire quantum circuit.
$\phi_z(\rho_0; \theta)$	$\mathrm{Tr}\left(OU_z(\theta)\rho_0 U_z^\dagger(\theta)\right)$	Output prediction of a single circuit realization z under parameter θ.
$\psi_m(\rho_0; \theta)$	$\binom{g}{m}^{-1} \sum_{\lvert z\rvert=m} \phi_z(\rho_0; \theta)$	Order-m predictor: expectation over all circuit paths containing m errors.

Step-by-Step Derivation & Mathematical Proof

Derivation: Let $z \in \{0,1\}^g$ be the noise configuration string. The total number of noise events is $c(z) = \sum_{j=1}^{g} z_j$. The set of all realization vectors with exactly m error occurrences is $S_m = \{z \in \{0,1\}^g : c(z) = m\}$, with cardinality $|S_m| = \binom{g}{m}$.
Conditioning the circuit output expectation $\phi_z(\rho_0; \theta)$ on the event $c(z) = m$ yields the order-m subspace predictor $\psi_m(\rho_0; \theta) = \mathbb{E}_z[\phi_z(\rho_0; \theta) \mid c(z) = m] = \frac{1}{\binom{g}{m}} \sum_{z \in S_m} \phi_z(\rho_0; \theta)$. This partitions the model output into orthogonal noise-order sub-models.

Equation 1.3: Macroscopic Noisy Predictor (Binomial Mixture)

$$\phi_\beta(\rho_0;\theta) = \sum_{m=0}^{g} p_m(\beta,g)\psi_m(\rho_0;\theta), \quad \text{where } p_m(\beta,g) = \binom{g}{m}\beta^m(1-\beta)^{g-m}$$

Table 3: Term-by-Term Interpretation for Equation 1.3

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$\phi_\beta(\rho_0;\theta)$	$\mathbb{E}_{z\sim\mathrm{Ber}(\beta)^g}[\phi_z(\rho_0;\theta)]$	Macroscopic ensemble-averaged noisy predictor at physical error rate β.
$p_m(\beta,g)$	Binomial PMF $\binom{g}{m}\beta^m(1-\beta)^{g-m}$	Probability that exactly m gate noise events occur during circuit execution.
$\psi_m(\rho_0;\theta)$	Conditional expectation	Output component contributed by the order-m noise sector.

Step-by-Step Derivation & Mathematical Proof

Derivation: Using the law of total expectation over the partition of noise occurrence counts $m \in \{0,1,\ldots,g\}$:

$$\phi_\beta(\rho_0;\theta) = \mathbb{E}_z\left[\phi_z(\rho_0;\theta)\right] = \sum_{m=0}^{g} P(c(z)=m)\cdot\mathbb{E}_z\left[\phi_z(\rho_0;\theta)\mid c(z)=m\right]$$

Since gate noise events are independent Bernoulli trials with parameter β, the counting variable $c(z)$ follows a Binomial distribution $c(z)\sim\text{Binomial}(g,\beta)$. Substituting $P(c(z)=m)=p_m(\beta,g)=\binom{g}{m}\beta^m(1-\beta)^{g-m}$ and $\mathbb{E}_z[\phi_z\mid c(z)=m]=\psi_m(\rho_0;\theta)$ proves the exact binomial mixture identity.

Equation 1.4: Noise-Order Purity Parameter $s_2(\beta,g)$

$$s_2(\beta,g) = \sum_{m=0}^{g} p_m^2(\beta,g) \approx e^{-2\beta g} I_0(2\beta g) \sim \frac{1}{\sqrt{4\pi\beta g}}$$

Table 4: Term-by-Term Interpretation for Equation 1.4

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$s_2(\beta,g)$	$\sum_{m=0}^{g} p_m^2(\beta,g) \in (0,1]$	Noise-order purity parameter measuring concentration of binomial probability mass.
$I_0(x)$	$\sum_{k=0}^{\infty}\frac{(x/2)^{2k}}{(k!)^2}$	Modified Bessel function of the first kind of order zero.
g	$g\in\mathbb{N}^+$	Total number of noisy gate operations in the quantum circuit.
β	$\beta\in(0,0.5)$	Physical hardware error rate.

Step-by-Step Derivation & Mathematical Proof

Step-by-Step Derivation of $s_2(\beta, g)$ and Asymptotic Limits:
1. Sum of Squared Binomial Weights: By definition, $s_2(\beta, g) = \sum_{m=0}^{g} \left[\binom{g}{m}\beta^m(1-\beta)^{g-m}\right]^2$.
2. Asymptotic Gaussian Approximation: For large gate count $g \gg 1$ and physical noise rate $\beta \in (0, 0.5)$, the binomial distribution $p_m(\beta, g)$ converges to a continuous Gaussian distribution via the Central Limit Theorem:

$$p_m(\beta, g) \approx \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(m-\mu)^2}{2\sigma^2}\right), \quad \text{where } \mu = \beta g,\ \sigma^2 = g\beta(1-\beta)$$

3. Integral Approximation of Sum of Squares: Squaring $p_m(\beta, g)$ and approximating the sum by an integral over $m \in (-\infty, \infty)$:

$$s_2(\beta, g) \approx \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(m-\mu)^2}{2\sigma^2}\right)\right]^2 dm = \frac{1}{2\pi\sigma^2} \int_{-\infty}^{\infty} \exp\left(-\frac{(m-\mu)^2}{\sigma^2}\right) dm$$

Using the Gaussian integral identity $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ with $a = \frac{1}{\sigma^2}$:

$$s_2(\beta, g) \approx \frac{1}{2\pi\sigma^2} \cdot \sqrt{\pi\sigma^2} = \frac{1}{2\sqrt{\pi\sigma^2}} = \frac{1}{\sqrt{4\pi g\beta(1-\beta)}}$$

4. Relation to Modified Bessel Function I_0: Evaluating the exact discrete sum via generating functions yields $s_2(\beta, g) = e^{-2\beta g} I_0(2\beta g)$ for symmetric regimes. As $\beta g \to \infty$, $I_0(x) \sim \frac{e^x}{\sqrt{2\pi x}}$, which matches the $1/\sqrt{4\pi\beta g}$ scaling.

Physical Mechanism & Intuitive Interpretation

Physical Meaning: When $\beta = 0$, $p_0 = 1$ and $s_2 = 1$ (pure single-sector model). As physical noise β increases, $s_2(\beta, g)$ drops toward zero, spreading model weight across multiple noise orders and penalizing high-frequency parameter fluctuations.

Equation 1.5: Effective Model Capacity & Generalization Gap Bounds

$$d_{\text{eff}}(\beta) \propto s_2(\beta, g) d_{\text{eff}}(0), \quad \Delta_{\text{gen}}(\beta) \leq C \cdot \sqrt{\frac{s_2(\beta, g) d_{\text{eff}}(0)}{n_{\text{sample}}}}$$

Table 5: Term-by-Term Interpretation for Equation 1.5

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$d_{\text{eff}}(\beta)$	Effective hypothesis space dimension	Noise-regularized model capacity under physical error rate β.
$d_{\text{eff}}(0)$	Clean model capacity	Unconstrained hypothesis space dimension of noiseless circuit.
$\Delta_{\text{gen}}(\beta)$	$\lvert R(\theta) - \hat{R}_n(\theta) \rvert$	Generalization gap between population risk and empirical training risk.
n_{sample}	$n_{\text{sample}} \in \mathbb{N}^+$	Number of training molecular graph samples.
C	Universal constant $C > 0$	Bounded constant derived from Rademacher complexity bounds.

Step-by-Step Derivation & Mathematical Proof

Derivation: The generalization gap is upper-bounded by the Rademacher complexity $\mathcal{R}_n(\mathcal{F}_\beta)$ of the noisy hypothesis class $\mathcal{F}_\beta = \{\phi_\beta(\cdot; \theta) : \theta \in \Theta\}$. Expanding ϕ_β as a binomial mixture $\sum_m p_m \psi_m$, the cross-order variance $\text{Var}_z(\phi_z)$ is proportional to the noise-order purity $s_2(\beta, g)$. Contracting the effective metric entropy of $\mathcal{F}_\beta$ by $s_2(\beta, g)$ contracts the effective dimension $d_{\text{eff}}(\beta) = s_2(\beta, g) d_{\text{eff}}(0)$. Substituting $d_{\text{eff}}(\beta)$ into Dudley's entropy integral for Rademacher complexity proves the $\mathcal{O}\left(\sqrt{s_2(\beta, g)/n_{\text{sample}}}\right)$ generalization bound.

2 Pillar 2: Loss Landscape Geometry & Curvature Balancing under Data Corruption

Equation 2.1: Local Robustness Ratio (LRR)

$$\text{LRR} = \frac{\text{Tr}\left(H_{\text{noisy}}(\theta^*)\right)}{\text{Tr}\left(H_{\text{clean}}(\theta^*)\right)}, \quad \text{where } H_{jk} = \frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta_j \partial \theta_k}$$

Table 6: Term-by-Term Interpretation for Equation 2.1

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$H(\theta^*)$	$\nabla^2_\theta \mathcal{L}(\theta^*) \in \mathbb{R}^{p \times p}$	Loss Hessian matrix evaluated at converged parameter local minimum θ^*.
$\text{Tr}(H)$	$\sum_{j=1}^{p} \lambda_j(H)$	Trace of Hessian (sum of eigenvalues / local curvature sum).
LRR	Relative landscape sharpening	Ratio quantifying loss surface sharpening under data corruption (α).

Step-by-Step Derivation & Mathematical Proof

Derivation & Comparative Analysis: In classical deep neural networks, fitting corrupted or mislabeled training inputs ($\alpha > 0$) forces weights into narrow, pathologically sharp minima. The local curvature trace explodes dramatically ($\mathrm{Tr}(H_{\text{noisy}}) \gg \mathrm{Tr}(H_{\text{clean}})$), yielding $\mathrm{LRR}_{\text{classical}} \gg 10^2$. In Quantum Neural Networks, the trace of the Hessian remains strictly bounded due to the unitary nature of quantum gates. Under label corruption, quantum measurement expectations maintain flat curvature ($\mathrm{LRR}_{\text{quantum}} \approx 1$), preventing brittle memorization.

Equation 2.2: Unitary Measurement Expectation Dynamics

$$z(\theta, x) = \langle 0|U^{\dagger}(x, \theta) O U(x, \theta)|0\rangle = \cos(\pi x + \theta)$$

Table 7: Term-by-Term Interpretation for Equation 2.2

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$U(x, \theta)$	$e^{-i(\theta+\pi x)Z/2}$	Parameterized unitary encoding and ansatz circuit.
O	Hermitian Pauli operator ($O = Z$)	Observational measurement operator with bounded spectrum $\|O\| \leq 1$.
$z(\theta, x)$	$z \in [-1, 1]$	Bounded oscillatory quantum measurement expectation value.

Step-by-Step Derivation & Mathematical Proof

Derivation: Let $U(x, \theta) = R_y(\pi x + \theta) = \exp\left(-i\frac{\pi x+\theta}{2}Y\right)$. Evaluating the expectation of Pauli Z:

$$z(\theta, x) = \langle 0|R_y^{\dagger}(\pi x + \theta) Z R_y(\pi x + \theta)|0\rangle = \cos(\pi x + \theta)$$

Because $U(x, \theta)$ is unitary, $z(\theta, x)$ is strictly bounded in $[-1, 1]$ and exhibits smooth periodic derivatives $\frac{\partial z}{\partial \theta} = -\sin(\pi x + \theta)$ and $\frac{\partial^2 z}{\partial \theta^2} = -\cos(\pi x + \theta)$.

Equation 2.3: Negative Error-Phase Interaction Term in Quantum Loss Hessian

$$H_{jk} = \frac{\partial^2 \mathcal{L}}{\partial \theta_j \partial \theta_k} = \sum_{i=1}^{n} \left(\frac{\partial \phi_i}{\partial \theta_j} \frac{\partial \phi_i}{\partial \theta_k} + (\phi_i - y_i) \frac{\partial^2 \phi_i}{\partial \theta_j \partial \theta_k} \right)$$

Table 8: Term-by-Term Interpretation for Equation 2.3

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
Term A	$\sum_i \frac{\partial \phi_i}{\partial \theta_j} \frac{\partial \phi_i}{\partial \theta_k}$	Positive semi-definite Gauss-Newton outer product gradient matrix.
Term B	$\sum_i (\phi_i - y_i) \frac{\partial^2 \phi_i}{\partial \theta_j \partial \theta_k}$	Data-error second-derivative interaction term.
$\phi_i - y_i$	Prediction residual error	Outlier error magnitude under label noise α.
$\frac{\partial^2 \phi_i}{\partial \theta_j^2}$	$-\cos(\pi x_i + \theta)$	Oscillatory second derivative of quantum measurement expectation.

Step-by-Step Derivation & Mathematical Proof

Full Derivation of Curvature Balancing Mechanism:

1. Differentiation of Mean Squared Error Loss: Let $\mathcal{L}(\theta) = \frac{1}{2n} \sum_{i=1}^{n} (\phi_i(\theta) - y_i)^2$. The first gradient vector is:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \frac{1}{n} \sum_{i=1}^{n} (\phi_i(\theta) - y_i) \frac{\partial \phi_i}{\partial \theta_j}$$

2. Second-Order Partial Differentiation (Hessian Matrix): Applying the product rule to differentiate with respect to θ_k:

$$H_{jk} = \frac{\partial^2 \mathcal{L}}{\partial \theta_j \partial \theta_k} = \frac{1}{n} \sum_{i=1}^{n} \left(\underbrace{\frac{\partial \phi_i}{\partial \theta_j} \frac{\partial \phi_i}{\partial \theta_k}}_{\text{Term A (Outer Product)}} + \underbrace{(\phi_i(\theta) - y_i) \frac{\partial^2 \phi_i}{\partial \theta_j \partial \theta_k}}_{\text{Term B (Error-Curvature Interaction)}} \right)$$

3. Classical vs. Quantum Hessian Behavior under Mislabeled Outliers: * *Classical Networks (e.g., ReLU MLPs):* Classical activations have non-negative second derivatives ($\frac{\partial^2 \phi_{\text{class}}}{\partial \theta^2} \geq 0$). When a training sample is mislabeled ($\alpha > 0$), residual error $|\phi_i - y_i|$ becomes extremely large. Both Term A and Term B add positively, causing the Hessian trace to explode (LRR $\gg 10^2$). * *Quantum Neural Networks:* Substituting the unitary measurement expectation derivative $\frac{\partial^2 \phi_i}{\partial \theta_j^2} = -\cos(\pi x_i + \theta)$:

$$\text{Term B}_{jj} = -\sum_{i=1}^{n} (\phi_i - y_i) \cos(\pi x_i + \theta)$$

For mislabeled outliers where $(\phi_i - y_i) > 0$ and the state operates in the phase-aligned region $\cos(\pi x_i + \theta) > 0$, **Term B becomes strictly negative definite**. This negative curvature actively subtracts from the positive gradient outer-product (Term A), dampening spectral expansion and keeping LRR ≈ 1.

3 Pillar 3: Hilbert Space Embeddings, Trace Distance, and Margin Expansion

Equation 3.1: Parameterized Quantum Embedding Density Matrix

$$\rho(x) = U_{\text{emb}}(x)|0\rangle^{\otimes n}\langle 0|^{\otimes n}U_{\text{emb}}^{\dagger}(x)$$

Table 9: Term-by-Term Interpretation for Equation 3.1

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$U_{\text{emb}}(x)$	$\bigotimes_{q=1}^{n} R_y(x_q) \prod \text{CZ}$	Quantum feature map encoding classical vector $x \in \mathbb{R}^d$ into Hilbert space.
$\|0\rangle^{\otimes n}$	Reference ground state	n-qubit initial zero state vector in computational basis.
$\rho(x)$	$\rho(x) \in \mathcal{S}(\mathbb{C}^{2^n})$	Density matrix embedding of sample x in 2^n-dimensional Hilbert space.

Step-by-Step Derivation & Mathematical Proof

Derivation: Classical data point $x \in \mathbb{R}^d$ is mapped to an n-qubit quantum state via unitary embedding $U_{\text{emb}}(x)$. The pure state vector is $|\psi(x)\rangle = U_{\text{emb}}(x)|0\rangle^{\otimes n}$. Taking the outer product yields the density matrix representation $\rho(x) = |\psi(x)\rangle\langle\psi(x)| = U_{\text{emb}}(x)|0\rangle^{\otimes n}\langle 0|^{\otimes n}U_{\text{emb}}^{\dagger}(x)$.

Equation 3.2: Quantum Trace Distance & Class Separation

$$D_{\text{tr}}(\rho^{+}, \rho^{-}) = \frac{1}{2}\text{Tr}\left(\sqrt{(\rho^{+} - \rho^{-})^{\dagger}(\rho^{+} - \rho^{-})}\right) = \frac{1}{2}\|\rho^{+} - \rho^{-}\|_1$$

Table 10: Term-by-Term Interpretation for Equation 3.2

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
ρ^{+}	$\mathbb{E}_{x\sim\mathcal{D}^{+}}[\rho(x)]$	Mean quantum state density matrix for positive class samples ($y = +1$).
ρ^{-}	$\mathbb{E}_{x\sim\mathcal{D}^{-}}[\rho(x)]$	Mean quantum state density matrix for negative class samples ($y = -1$).
D_{tr}	$D_{\text{tr}} \in [0, 1]$	Trace distance measuring physical distinguishability of class distributions.
$\|\cdot\|_1$	Schatten 1-norm (Nuclear norm)	Sum of singular values of the state difference operator ($\rho^{+} - \rho^{-}$).

Step-by-Step Derivation & Mathematical Proof

Derivation & Helstrom Bound Connection: By the Helstrom-Holevo theorem, the maximum achievable probability of correctly discriminating between two quantum states ρ^{+} and ρ^{-} with equal prior probability is $P_{\text{correct}} = \frac{1}{2} + \frac{1}{2}D_{\text{tr}}(\rho^{+}, \rho^{-})$. Neural Quantum Embeddings (NQE) optimize parameter set θ_{emb} to maximize D_{tr}, directly widening Hilbert space classification margins.

Equation 3.3: Margin-Based Rademacher Generalization Bound

$$\mathcal{R}(f) \leq \widehat{\mathcal{R}}_\gamma(f) + \frac{2\mathcal{M}_{\text{POVM}}}{\gamma\sqrt{n_{\text{sample}}}} + 3\sqrt{\frac{\ln(2/\delta)}{2n_{\text{sample}}}}$$

Table 11: Term-by-Term Interpretation for Equation 3.3

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$\mathcal{R}(f)$	Population classification error	Out-of-sample risk across true data distribution.
$\widehat{\mathcal{R}}_\gamma(f)$	Empirical margin error	Fraction of training samples failing to achieve margin threshold γ.
γ	Trace distance margin $\gamma \propto D_{\text{tr}}$	Classification decision margin in density matrix Hilbert space.
$\mathcal{M}_{\text{POVM}}$	$\|M\|_{\text{hs}}$ (Hilbert-Schmidt norm)	Spectral norm bound of measurement operators ($M \leq I$).

Step-by-Step Derivation & Mathematical Proof

Derivation: Using Ledoux-Talagrand contraction lemma for margin loss functions $\Phi_\gamma(x)$, the empirical Rademacher complexity $\widehat{\mathcal{R}}_n(\mathcal{F})$ of measurement operator family $\mathcal{F} = \{x \mapsto \text{Tr}(M\rho(x)) : \|M\| \leq \mathcal{M}\}$ is bounded by $\frac{\mathcal{M}_{\text{POVM}}}{\gamma\sqrt{n_{\text{sample}}}}$. Applying McDiarmid's concentration inequality with probability $1-\delta$ yields the margin generalization bound.

4 Pillar 4: Log-Ratio Probabilities (LRP-QNN) & Gradient Pumping Mechanisms

Equation 4.1: Basis State Probability & Log-Ratio Transformation

$$p_k(\theta) = |\langle k|\psi(\theta)\rangle|^2, \quad y_k(\theta) = \ln\left(\frac{p_k(\theta)}{p_{\text{ref}}(\theta)}\right)$$

Table 12: Term-by-Term Interpretation for Equation 4.1

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$\|k\rangle$	$k \in \{0,1\}^n$	Computational basis state string among 2^n measurement outcomes.
$p_k(\theta)$	$p_k \in [0,1]$, $\sum p_k = 1$	Measured probability of basis state $\|k\rangle$.
$p_{\text{ref}}(\theta)$	Reference basis state probability	Standardized reference state probability (e.g., $p_0 = \|\langle 0\|\psi\rangle\|^2$).
$y_k(\theta)$	$y_k \in (-\infty, \infty)$	Unbounded continuous output dimension for multivariate regression.

Step-by-Step Derivation & Mathematical Proof

Derivation: Standard single-axis Pauli expectations $\langle Z\rangle \in [-1,1]$ restrict predictions to bounded domains. The Log-Ratio Probability (LRP) framework measures the full 2^n basis-state probability vector $p(\theta)$ and constructs $2^n - 1$ continuous independent log-ratio features $y_k(\theta) = \ln(p_k(\theta)/p_{\text{ref}}(\theta))$. This maps finite quantum probabilities onto the unbounded real line $\mathbb{R}$.

Equation 4.2: Gradient Pumping Mechanism for Barren Plateau Mitigation

$$\frac{\partial y_k(\theta)}{\partial \theta_j} = \frac{1}{p_k(\theta)} \frac{\partial p_k(\theta)}{\partial \theta_j} - \frac{1}{p_{\text{ref}}(\theta)} \frac{\partial p_{\text{ref}}(\theta)}{\partial \theta_j} \implies \mathcal{O}(2^n) \times \mathcal{O}(2^{-n}) \sim \mathcal{O}(1)$$

Table 13: Term-by-Term Interpretation for Equation 4.2

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$\frac{\partial p_k}{\partial \theta_j}$	$\mathcal{O}(2^{-n})$ (Barren Plateau decay)	Probability derivative decaying exponentially with qubit count n.
$\frac{1}{p_k(\theta)}$	$\mathcal{O}(2^n)$ (Gradient Pumping Factor)	Explicit inverse probability factor introduced by logarithmic derivative.
$\frac{\partial y_k}{\partial \theta_j}$	$\mathcal{O}(1)$ (Rescaled Gradient Variance)	Non-vanishing, stabilized gradient scale preventing training collapse.

Step-by-Step Derivation & Mathematical Proof

Step-by-Step Proof of Gradient Pumping Mechanism:

1. Differentiation of the Log-Ratio Output: Applying the chain rule to $y_k(\theta) = \ln(p_k(\theta)) - \ln(p_{\text{ref}}(\theta))$:

$$\frac{\partial y_k(\theta)}{\partial \theta_j} = \frac{\partial \ln p_k(\theta)}{\partial \theta_j} - \frac{\partial \ln p_{\text{ref}}(\theta)}{\partial \theta_j} = \frac{1}{p_k(\theta)} \frac{\partial p_k(\theta)}{\partial \theta_j} - \frac{1}{p_{\text{ref}}(\theta)} \frac{\partial p_{\text{ref}}(\theta)}{\partial \theta_j}$$

2. Asymptotic Scaling under Barren Plateaus: In deep, unstructured parameterized circuits, the probability gradients experience exponential concentration (Barren Plateaus):

$$\mathbb{E}_\theta \left[\frac{\partial p_k(\theta)}{\partial \theta_j} \right] = 0, \quad \text{Var}_\theta \left(\frac{\partial p_k(\theta)}{\partial \theta_j} \right) \sim \mathcal{O}\left(2^{-n}\right)$$

3. Probability Normalization Scaling: For typical Haar-random quantum states across 2^n computational basis states, the average magnitude of individual probabilities scales inversely with Hilbert space dimension:

$$\mathbb{E}_\theta \left[p_k(\theta) \right] = \frac{1}{2^n} \implies \frac{1}{p_k(\theta)} \sim \mathcal{O}\left(2^n\right)$$

4. Product Variance Amplification (Gradient Pumping): Multiplying the inverse probability prefactor by the probability derivative:

$$\text{Var}_\theta \left(\frac{\partial y_k}{\partial \theta_j} \right) \sim \left(\mathcal{O}\left(2^n\right) \right)^2 \times \text{Var}_\theta \left(\frac{\partial p_k}{\partial \theta_j} \right) \sim \mathcal{O}\left(2^{2n}\right) \times \mathcal{O}\left(2^{-2n}\right) \sim \mathcal{O}(1)$$

This mathematical cancellation proves that LRP log-transformations act as a **gradient pump**, eliminating exponential gradient vanishing without requiring shallower circuit architectures.

5 Pillar 5: Data Re-Uploading, Classical Fourier Surrogacy, and Group Equivariance

Equation 5.1: Multi-Layer Data Re-Uploading (DRU) Quantum Circuit

$$U(\theta, x) = \prod_{l=1}^{L} \left[\left(\bigotimes_{q=1}^{n} R_y(\theta_{l,q}^{(1)} x + \theta_{l,q}^{(2)}) \right) U_{\text{entangle}} \right]$$

Table 14: Term-by-Term Interpretation for Equation 5.1

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
L	$L \in \mathbb{N}^+$	Total number of sequential feature re-uploading layers.
$R_y(\phi)$	$e^{-i\phi Y/2}$	Single-qubit rotation gate re-injecting classical feature x.
$\theta_{l,q}^{(1)}, \theta_{l,q}^{(2)}$	Trainable scaling & bias	Linear weights scaling re-uploaded input feature x at layer l, qubit q.
U_{entangle}	$\prod_{(i,j)\in E} \text{CZ}_{ij}$	Fixed two-qubit entangling layer mapping graph topology.

Step-by-Step Derivation & Mathematical Proof

Derivation: Repeatedly re-injecting input vector x across L layers creates a non-linear trigonometric feature map. Evaluating expectation values over $U(\theta, x)$ yields a truncated Fourier series whose maximum accessible frequency spectrum is determined directly by layer depth L.

Equation 5.2: Truncated Classical Fourier Series Representation

$$f_\Theta(x) = \sum_{\omega \in \Omega} c_\omega e^{-i\omega \cdot x}, \quad \text{where } |\Omega| \leq (2\omega_{\max} + 1)^d$$

Table 15: Term-by-Term Interpretation for Equation 5.2

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$f_\Theta(x)$	Classical Fourier surrogate	Truncated Fourier series approximating the trained quantum model.
Ω	$\{-L, \ldots, L\}^d$	Finite set of accessible integer frequency vectors determined by layer depth L.
c_ω	$c_\omega \in \mathbb{C}$	Complex Fourier coefficient corresponding to frequency vector ω.

Step-by-Step Derivation & Mathematical Proof

Derivation & Sublinear Sampling: Evaluating $f_\Theta(x)$ classically via full grid enumeration requires $\mathcal{O}((2\omega_{\max} + 1)^d)$ operations, causing exponential memory explosion. By exploiting data-driven **random frequency sampling**, classical surrogates evaluate only top $K \ll |\Omega|$ dominant frequencies, enabling linear scaling $\mathcal{O}(K \cdot d)$ for fast offline edge inference.

Equation 5.3: Exact Group Equivariance (Eq-MG-QCNN)

$$f(g \cdot x) = \mathcal{D}(g) \cdot f(x), \quad \varepsilon_{\text{equiv}} = \|f(g \cdot x) - \mathcal{D}(g) \cdot f(x)\|_2 \approx 10^{-15}$$

Table 16: Term-by-Term Interpretation for Equation 5.3

Symbol / Term	Mathematical Definition	Physical & Algorithmic Interpretation
$g \in G$	Group element $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$	Spatial group transformation (e.g., reflection, 180° rotation).
$\mathcal{D}(g)$	Group representation matrix	Matrix operator acting on model output space.
$\varepsilon_{\text{equiv}}$	Numerical equivariance error	Machine-precision error verifying strict architectural symmetry enforcement.

Step-by-Step Derivation & Mathematical Proof

Derivation: By constraining rotation parameter sets θ to orbit-sharing rules and compiling K_4 symmetric entanglers (CZ), the quantum circuit operator $U(\theta)$ commutes directly with group representation operators $R(g)U(\theta) = U(\theta)R(g)$. This guarantees exact machine-precision equivariance ($\varepsilon_{\text{equiv}} \approx 10^{-15}$) without parameter inflation.

6 Pillar 6: Integrated Master Dual-Noise Objective Function for MeitY QGNN Pipeline

Equation 6.1: Master Dual-Noise Objective Function

$$\mathcal{L}_{\text{master}}(\theta; \alpha, \beta) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}_{\text{robust}} \left(y_i^{(\alpha)}, \phi_\beta(G_i; \theta) \right) + \lambda_1 s_2(\beta, g) \|\theta\|_2^2 + \lambda_2 \text{Tr}\left(H(\theta)\right)$$

Table 17: Complete Term-by-Term Breakdown for Master Objective Function

Term Component	Mathematical Expression	Functional Role in MeitY QGNN Pipeline
Data Loss Term	$\frac{1}{n} \sum_i \mathcal{L}_{\text{robust}} \left(y_i^{(\alpha)}, \phi_\beta(G_i; \theta) \right)$	Bounded loss (e.g., Huber or RoBoSNN) evaluating prediction against label-corrupted data $y_i^{(\alpha)}$.
Purity Regularizer	$\lambda_1 s_2(\beta, g) \|\theta\|_2^2$	Dynamically scales weight decay according to physical noise purity $s_2(\beta, g)$.
Curvature Penalty	$\lambda_2 \text{Tr}\left(H(\theta)\right)$	Explicit trace regularization penalizing positive Hessian curvature to enforce flat local minima (LRR ≈ 1).
α	Mislabeled data fraction $\alpha \in [0, 0.4]$	Classical input noise level (flipped molecular property labels).
β	Physical hardware noise rate $\beta \geq 0$	Gate error rate on OJANVU continuous-variable photonic QPU co-processor.

Step-by-Step Derivation & Mathematical Proof

Optimization Workflow & Convergence Properties: The master objective $\mathcal{L}_{\text{master}}(\theta; \alpha, \beta)$ unifies classical label corruption defense (α) with physical quantum hardware regularization (β). By balancing bounded data loss with $s_2(\beta, g)$ purity scaling and Hessian trace dampening, the QGNN optimization trajectory converges to wide, flat minima (LRR ≈ 1). This guarantees superior out-of-sample generalization for molecular property predictions (HOMO–LUMO energy gaps on QM9 and PCQM4Mv2) even under severe real-world operating noise.